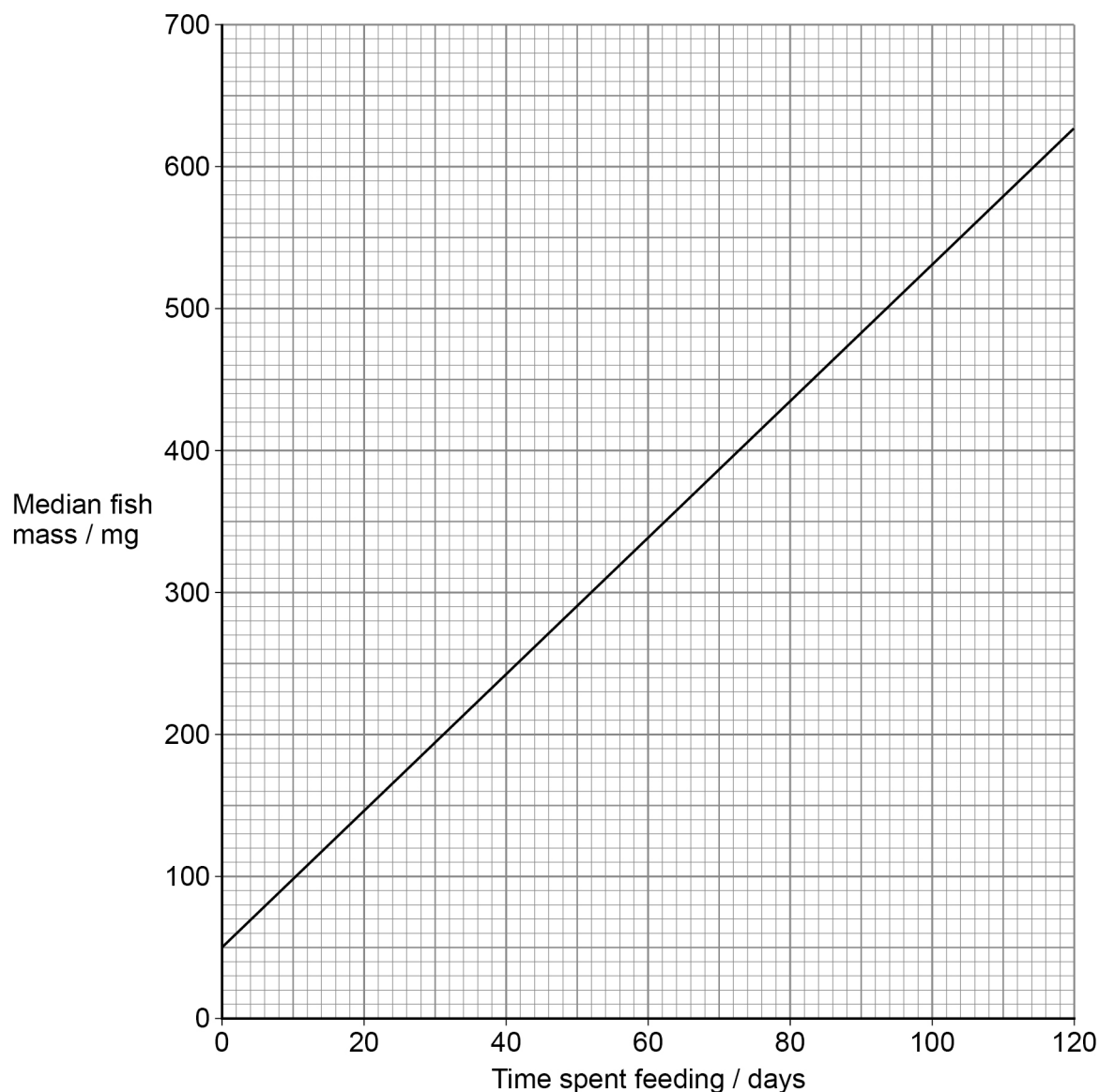


0 6

Trout is a type of fish, often produced commercially in trout farms.

A scientist investigated the growth of farmed trout. She determined the median mass of a large population of trout at intervals. She started measuring on the day the newly hatched fish began feeding. Her results are shown in **Figure 4**.

Figure 4



The best fit line shown in **Figure 4** is represented using this equation.

$$\text{median fish mass} = (m \times \text{days feeding}) + 50$$

where m is the gradient of the best fit line.

Do not write
outside the
box



0 6 . 1

Use **Figure 4** and the equation to calculate the median mass of fish after 195 days' feeding.

Show your working.

[2 marks]

Answer _____ mg

0 6 . 2

A trout body cell contains 80 chromosomes.

Table 2 shows the number of chromosomes and the mass of DNA in different nuclei. All the nuclei are from the same trout.

Complete **Table 2**.

[2 marks]

Table 2

Nucleus	Number of chromosomes	Mass of DNA / arbitrary units
At prophase of mitosis	80	
At telophase of mitosis		25
From an egg cell		

0 6 . 3

Give **one** reason why trout eggs produced by meiosis are genetically different.

[1 mark]

Question 6 continues on the next page

Turn over ►



A trout body cell contains 80 chromosomes.

Farmed female trout are treated so that they produce diploid egg cells.

0 6 . 4

Give the number of chromosomes in body cells of the offspring produced from treated farmed female trout and untreated farmed male trout.

[1 mark]

Number of chromosomes _____

0 6 . 5

The offspring produced from farmed trout are sterile. Suggest and explain why.

[2 marks]

8

